

Windows Environment — Complete Implementation Plan

Reconciliation of WSC2026_TP39 Topology and WSC2024_TP39 Module B Instructions

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Document Control & Basis of Plan

The two source attachments **do not describe the same environment**.

Attribute	WSC2026_TP39_topology_pre.pdf	WSC2024_TP39_MB_actual_en.pdf
Status	Pre-release, values redacted	Final, complete
Sites	Shanghai, Beijing	Paris, Lyon, Los Angeles
Zones / layers	Internet, DMZ, INTERNAL, CLIENT	Core, Distribution, Client, ISP
OS baseline	Windows Server 2025 (Core + GUI), Windows 11	Windows Server 2022 21H2, Windows 11 23H2
FQDNs / IPs / services	<redacted>	Fully specified

Basis of plan: the 2024 Module B document is the single source of truth for all hostnames, IP addresses, domain names, credentials and service requirements. The 2026 topology is used only as a forward-looking design overlay. Every value originating solely from the 2026 file, or absent from both files, is marked [TO BE CONFIRMED].

1. Environment Summary

1.1 Sites, Domains and Addressing

Element	Value	Source
Forest root domain	paris.local	Instructions, Task 1
Child domain	lyon.paris.local	Instructions, Task 3

Element	Value	Source
NetBIOS (root)	PARIS (implied by VPN credential Domain: PARIS)	Instructions, Task 5
Forest / domain functional level	[TO BE CONFIRMED] — not stated; use WinServer2016 minimum	—
AD Sites	[TO BE CONFIRMED] — not defined; recommend Paris and Lyon	—
Workgroup (ISP)	ISP on LA-ROUTER, non-domain-joined	Instructions, Task 4
VLAN IDs	[TO BE CONFIRMED] — no VLAN tags in either document; segmentation by subnet only	—
Subnet masks	Not stated; topology labels all as /24	Topology diagram

1.2 Subnets

Subnet	Purpose	Gateway	DHCP
10.10.0.0/24	Paris Core Layer	10.10.0.1 (PARIS-ROUTER)	Static
10.20.0.0/24	Paris Distribution Layer	10.20.0.1 (PARIS-ROUTER)	Static
10.30.0.0/24	Paris Client Network	10.30.0.1 (PARIS-ROUTER)	Yes — NW-SRV
10.40.0.0/24	Lyon Remote Network	10.40.0.1 (LYON-ROUTER)	Yes — LYON-ROUTER
20.0.0.0/24	LA Exchange to Paris transit	20.0.0.1 (LA-ROUTER)	Static
30.0.0.0/24	LA Exchange to Lyon transit	30.0.0.1 (LA-ROUTER)	Static

1.3 Node Inventory

Hostname	FQDN	Role	OS / Edition	IP(s)	Specs	Dependencies
DC1	dc1.paris.local	Forest root DC, primary DNS	WS2022 21H2, GUI	10.10.0.10	[TO BE CONFIRMED]	First build — none
NW-SRV	nw-srv.paris.local	Secondary DNS, DHCP (10.30.0.0/24)	WS2022 21H2, Core	10.10.0.11	[TO BE CONFIRMED]	DC1; PARIS-ROUTER (relay)
FILE-SRV	file-srv.paris.local	File Services, FSRM, DFS-N/R primary,	WS2022 21H2, GUI	10.20.0.10	[TO BE CONFIRMED]	DC1; WEB-SRV; PARIS-ROUTER

Hostname	FQDN	Role	OS / Edition	IP(s)	Specs	Dependencies
		RDS/RemoteApp				
WEB-SRV	web-srv.paris.local	AD CS Root CA, IIS, DFS-R target	WS2022 21H2, GUI	10.20.0.11	[TO BE CONFIRMED]	DC1; FILE-SRV
PARIS-ROUTER	paris-router.paris.local	RRAS routing, IKEv2 S2S VPN, IIS + ARR reverse proxy, DHCP relay	WS2022 21H2, GUI	10.10.0.1, 10.20.0.1, 10.30.0.1, 20.0.0.2	[TO BE CONFIRMED]	DC1; LA-ROUTER; WEB-SRV
LA-ROUTER	— (workgroup ISP)	RRAS routing, DNS for paris.com	WS2022 21H2, GUI	20.0.0.1, 30.0.0.1	[TO BE CONFIRMED]	None — standalone
LYON-ROUTER	lyon-router.lyon.paris.local	RRAS routing, IKEv2 S2S VPN, DHCP (10.40.0.0/24)	WS2022 21H2, GUI	10.40.0.1, 30.0.0.2	[TO BE CONFIRMED]	LA-ROUTER; DC2
DC2	dc2.lyon.paris.local	Child-domain DC, DNS	WS2022 21H2, GUI	10.40.0.10	[TO BE CONFIRMED]	DC1; LYON-ROUTER; VPN
WIN-CLIENT1	—	Paris client	Windows 11 23H2	DHCP (10.30.0.151–200)	[TO BE CONFIRMED]	NW-SRV DHCP; DC1
WIN-CLIENT2	—	Lyon client	Windows 11 23H2	DHCP (10.40.0.151–200)	[TO BE CONFIRMED]	LYON-ROUTER DHCP; DC2
ANSIBLE-SRV	— (workgroup)	Ansible control node, git, VS Code, Zeal	Debian	DHCP on 10.30.0.0/24	[TO BE CONFIRMED]	FILE-SRV, DC2, DC1 (WinRM)

Virtualization and storage: the topology renders every node as a VM icon, but hypervisor product, host count, storage platform, backup target, antivirus and monitoring are not specified in either document — all [TO BE CONFIRMED].

1.4 2026 Design Overlay (not buildable as-is)

2026 element	Value	Status
Sites	Shanghai, Beijing	Defined
Zones	Internet, DMZ, INTERNAL, CLIENT	Defined
OS baseline	Windows Server 2025 (Core + GUI), Windows 11	Defined
Client tooling	RSAT, WSL (Debian)	Defined
All FQDNs, IPs, services	<redacted>	[TO BE CONFIRMED]

2. Services and Roles Inventory

#	Role / feature / software	Host	Why required	Prerequisites	Edition / licence	Ports & protocols
1	AD DS (AD-Domain-Services) + RSAT-ADDS	DC1	Task 1: new forest paris.local	Static IP, correct hostname	Std or DC 2022	TCP 88, 135, 389, 445, 464, 636, 3268/3269, 49152–65535; UDP 88, 123, 389, 464
2	DNS Server (primary + reverse)	DC1	Task 1 DNS; AD locator records	AD DS	Included	TCP/UDP 53
3	DNS Server (secondary)	NW-SRV	Task 1: secondary of all forward + reverse zones	Zone transfer permitted on DC1	Included	TCP/UDP 53 (AXFR/IXFR on TCP)
4	DHCP Server	NW-SRV	Task 1: scope client for 10.30.0.0/24	Domain join + Add-DhcpServerInDC	Included	UDP 67/68; TCP 647
5	DHCP Relay Agent (RRAS)	PARIS-ROUTER	Gap fix: DHCP server on 10.10.0.0/24 serves 10.30.0.0/24 — broadcasts will not cross	RRAS installed	Included	UDP 67
6	File & Storage Services (FS-FileServer)	FILE-SRV, WEB-SRV	Group shares, UserShare, DFS targets	Domain join	Included	TCP 445, 139
7	FSRM (FS-Resource-Manager)	FILE-SRV	50 MB hard quota + executable file screen	FS-FileServer	Included	n/a (local)
8	DFS Namespaces (FS-DFS-Namespaces)	FILE-SRV	Namespace CSDrive, roots in C:\DFSRoots\	Domain join	Included	TCP 445, 389, 135
9	DFS Replication (FS-DFS-Replication)	FILE-SRV (primary), WEB-SRV	Replication for all DFS shares	DFS-N, both domain-joined	Included	TCP 135 + dynamic RPC
10	RDS — Session Host, Connection Broker, Web Access	FILE-SRV	Publish notepad.exe to Paris and Lyon	Domain join; RD Licensing [TO BE CONFIRMED]	RDS CALs beyond 120-day grace	TCP/UDP 3389; TCP 443, 5504, 5985

#	Role / feature / software	Host	Why required	Prerequisites	Edition / licence	Ports & protocols
11	AD CS — CA + Web Enrolment	WEB-SRV	Root CA for paris.local; certs for www, help	Domain join; Enterprise Admin	Std supports Enterprise CA	TCP 135 + dynamic RPC; TCP 80 (CRL/AIA)
12	IIS (Web-Server)	WEB-SRV	Default site www.paris.local; vhosts help., external.	—	Included	TCP 80, 443
13	IIS + ARR + URL Rewrite	PARIS-ROUTER	Reverse proxy for external site	requestRouter_amd64.msi, rewrite_amd64_en-US.msi on Admin Desktop	Free MS add-ons	TCP 80, 443
14	URL Rewrite	WEB-SRV	HTTP to HTTPS redirect for www, help	IIS	Free	—
15	Remote Access / RRAS (Routing)	PARIS-ROUTER, LA-ROUTER, LYON-ROUTER	Default gateways for all subnets	Two or more NICs, IP forwarding	Included	IP forwarding; UDP 500/4500
16	RRAS — Site-to-Site VPN (IKEv2)	PARIS-ROUTER and LYON-ROUTER	Task 5	Routing installed first	Included	UDP 500, 4500; ESP/50
17	DNS Server (standalone)	LA-ROUTER	Zone paris.com, record external to 20.0.0.2	Workgroup only — do not domain-join	Included	TCP/UDP 53
18	AD DS + DNS (child domain)	DC2	Child domain lyon.paris.local	Routed/VPN path to DC1; parent delegation	Std or DC	Same as row 1
19	DHCP Server	LYON-ROUTER	Scope client for 10.40.0.0/24	Domain join + authorisation	Included	UDP 67/68
20	RSAT (GPMC, AD, DNS, DHCP, DFS)	DC1 / admin workstation	GPO and directory administration	—	Included	—
21	Ansible + git + pywinrm	ANSIBLE-SRV (Debian)	Task 7 share automation	WinRM + Kerberos on targets	Open source	TCP 5985/5986; TCP/UDP 88, 464
22	Hypervisor	[TO BE CONFIRMED]	All nodes drawn as VMs	—	[TO BE CONFIRMED]	—

#	Role / feature / software	Host	Why required	Prerequisites	Edition / licence	Ports & protocols
23	Backup / antivirus / monitoring	Absent from both documents	—	—	—	—

3. Build Order & Dependency Map

Phase	Action	Why it must precede the next step	Parallel?
0	Hypervisor, VM provisioning, virtual switches per subnet, ISO/media staging	Nothing can be built without Layer 1/2	—
1	LA-ROUTER: OS, hostname, workgroup <code>ISP</code> , <code>20.0.0.1 / 30.0.0.1</code> , RRAS routing, DNS zone <code>paris.com</code>	LA-ROUTER is the transit core; Paris and Lyon have no path until it routes. Zero upstream dependencies	Yes — with Phase 2
2	DC1: OS, <code>10.10.0.10</code> , <code>Install-ADDSForest</code> (installs DNS)	DNS must exist before any domain join; all hosts depend on <code>paris.local</code> SRV records	Yes — with Phase 1
3	PARIS-ROUTER: four interfaces, join <code>paris.local</code> , RRAS routing, DHCP relay	Routing must exist before <code>10.20.x / 10.30.x</code> hosts reach DC1; relay must exist before Phase 5 DHCP works	—
4	Reverse zones + A records on DC1; zone-transfer permissions to <code>10.10.0.11</code>	Secondary zones on NW-SRV fail without transfer rights	—
5	NW-SRV: IP, domain join, secondary DNS zones, DHCP scope, authorisation	DHCP authorisation requires an existing domain; secondary zones require Phase 4	—
6a	FILE-SRV: domain join, FS + FSRM + DFS-N/R, shares, quotas, file screens	NTFS ACLs reference AD groups — groups must exist first	Yes — with 6b
6b	WEB-SRV: domain join, AD CS Enterprise Root CA first , then IIS	PKI before IIS: HTTPS bindings and the “no certificate errors” requirement need the root distributed via AD before certs are issued	Yes — with 6a
7	DFS namespace <code>CSDrive</code> , folder targets, DFS-R group FILE-SRV to WEB-SRV	Requires both file servers domain-joined and shares created	—

Phase	Action	Why it must precede the next step	Parallel?
8	OUs, groups, C:\create_user.ps1, users, HR fine-grained password policy, GPOs	GPO link targets are the OUs; Phase 6a ACLs reference these groups	Group creation parallel with Phase 6
9	LYON-ROUTER : 10.40.0.1 / 30.0.0.2, RRAS routing	Must route before DC2 can reach DC1	—
10	IKEv2 S2S VPN PARIS-ROUTER to LYON-ROUTER, persistent	Must precede DC2 promotion — child-domain promotion needs LDAP/Kerberos/RPC to DC1 across the tunnel	—
11	DC2 : Install-ADDSDomain child domain lyon.paris.local with DNS delegation	Needs Phase 10; delegation record must exist in the parent zone	—
12	LYON-ROUTER domain join, DHCP scope, authorisation	DHCP authorisation needs DC2 online	—
13	RemoteApp deployment on FILE-SRV, published notepad.exe	Needs AD and Lyon reachability to satisfy “accessible from both client networks”	—
14	WIN-CLIENT1 (paris.local), WIN-CLIENT2 (lyon.paris.local)	Need DHCP, DNS and domains live	Yes — with each other
15	GPO validation, drive mappings, certificate trust test in Edge	Requires clients	—
16	ANSIBLE-SRV : git init, manage-shares.yaml, credential-less run, final commit	Requires AD groups, FILE-SRV, DC2 and WinRM all live	—

4. Detailed Installation Procedure

4.0 Common Baseline — Every Windows Server

1. Boot the ISO and install **Windows Server 2022 Standard**. Choose **Desktop Experience (GUI)** for DC1, FILE-SRV, WEB-SRV, PARIS-ROUTER, LA-ROUTER, LYON-ROUTER and DC2. Choose **Server Core** for NW-SRV.
2. Set the local Administrator password to `skill139@Lyon`.
3. Set the time zone and confirm NTP. AD Kerberos fails beyond a five-minute clock skew.
4. Apply the hardening baseline below. This is not specified in either document — flag before deviating.

```
# Run on every Windows Server as the first post-install step
Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol -NoRestart
Set-TimeZone -Id "Romance Standard Time" # [TO BE CONFIRMED] - time zone not specified
Set-NetFirewallProfile -Profile Domain,Private,Public -Enabled True
```

Core hosts (NW-SRV): there is no Server Manager GUI. Use `sconfig` for basics and PowerShell for everything else, or manage remotely from DC1 via RSAT / Enter-PSsession.

4.1 DC1 — dc1.paris.local (10.10.0.10)

Network and naming

```
$if = (Get-NetAdapter | Where-Object Status -eq 'Up').Name
New-NetIPAddress -InterfaceAlias $if -IPAddress 10.10.0.10 -PrefixLength 24 -DefaultGateway 10.10.0.1
Set-DnsClientServerAddress -InterfaceAlias $if -ServerAddresses 127.0.0.1
Rename-Computer -NewName DC1 -Restart
```

Server Manager GUI path

1. **Manage > Add Roles and Features** > Role-based or feature-based installation.
2. Select **Active Directory Domain Services > Add Features > Install**.
3. Click the yellow notification flag > **Promote this server to a domain controller**.
4. Select **Add a new forest**; Root domain name `paris.local`.
5. Set the DSRM password to `Skill39@Lyon`; accept the DNS delegation warning.
6. **Install** and allow the automatic restart.

PowerShell path

```
Install-WindowsFeature AD-Domain-Services,DNS -IncludeManagementTools

Import-Module ADDSDeployment
Install-ADDSForest `
  -DomainName "paris.local" `
  -DomainNetbiosName "PARIS" `
  -ForestMode "WinThreshold" `
  -DomainMode "WinThreshold" `
  -InstallDns:$true `
  -DatabasePath "C:\Windows\NTDS" -LogPath "C:\Windows\NTDS" -SysvolPath "C:\Windows\SYSVOL" `
  -SafeModeAdministratorPassword (ConvertTo-SecureString "Skill39@Lyon" -AsPlainText -Force) `
  -NoRebootOnCompletion:$false -Force
```

Post-install DNS

```
# Reverse zones - a reverse zone is required on the primary DNS server
Add-DnsServerPrimaryZone -NetworkId "10.10.0.0/24" -ReplicationScope Forest
Add-DnsServerPrimaryZone -NetworkId "10.20.0.0/24" -ReplicationScope Forest
Add-DnsServerPrimaryZone -NetworkId "10.30.0.0/24" -ReplicationScope Forest

# A records - Task 1 DNS table
Add-DnsServerResourceRecordA -ZoneName paris.local -Name "www" -IPv4Address 10.20.0.11 -CreatePtr
```

```
Add-DnsServerResourceRecordA -ZoneName paris.local -Name "help" -IPv4Address 10.20.0.11 -CreatePtr # [TO BE CONFIRMED]
Add-DnsServerResourceRecordA -ZoneName paris.local -Name "external" -IPv4Address 10.20.0.11 -CreatePtr # [TO BE CONFIRMED]
Add-DnsServerResourceRecordA -ZoneName paris.local -Name "app" -IPv4Address 10.20.0.10 -CreatePtr

# Permit zone transfer to the secondary (NW-SRV)
Set-DnsServerPrimaryZone -Name "paris.local" -SecureSecondaries TransferToSecureServers `
    -SecondaryServers 10.10.0.11 -Notify NotifyServers -NotifyServers 10.10.0.11
Set-DnsServerPrimaryZone -Name "0.10.10.in-addr.arpa" -SecureSecondaries TransferToSecureServers -SecondaryServers 10.10.0.11
Set-DnsServerPrimaryZone -Name "0.20.10.in-addr.arpa" -SecureSecondaries TransferToSecureServers -SecondaryServers 10.10.0.11
Set-DnsServerPrimaryZone -Name "0.30.10.in-addr.arpa" -SecureSecondaries TransferToSecureServers -SecondaryServers 10.10.0.11

# Internet forwarder - not specified in either document
Set-DnsServerForwarder -IPAddress 20.0.0.1 -PassThru # [TO BE CONFIRMED]
```

Warning: the instruction table lists `external` and `help` with no IP address — the cell is blank or merged under the `www` value of 10.20.0.11. Both are IIS vhosts on WEB-SRV, so 10.20.0.11 is the only coherent value. Confirm before build.

4.2 NW-SRV — nw-srv.paris.local (10.10.0.11, Server Core)

```
$if = (Get-NetAdapter | Where-Object Status -eq 'Up').Name
New-NetIPAddress -InterfaceAlias $if -IPAddress 10.10.0.11 -PrefixLength 24 -DefaultGateway 10.10.0.1
Set-DnsClientServerAddress -InterfaceAlias $if -ServerAddresses 10.10.0.10
Rename-Computer -NewName NW-SRV -Restart

# after reboot
Add-Computer -DomainName paris.local -Credential (Get-Credential PARIS\Administrator) -Restart
```

Secondary DNS

```
Install-WindowsFeature DNS -IncludeManagementTools

Add-DnsServerSecondaryZone -Name "paris.local" -ZoneFile "paris.local.dns" -MasterServers 10.10.0.10
Add-DnsServerSecondaryZone -Name "0.10.10.in-addr.arpa" -ZoneFile "0.10.10.in-addr.arpa.dns" -MasterServers 10.10.0.10
Add-DnsServerSecondaryZone -Name "0.20.10.in-addr.arpa" -ZoneFile "0.20.10.in-addr.arpa.dns" -MasterServers 10.10.0.10
Add-DnsServerSecondaryZone -Name "0.30.10.in-addr.arpa" -ZoneFile "0.30.10.in-addr.arpa.dns" -MasterServers 10.10.0.10

Start-DnsServerZoneTransfer -Name "paris.local" -FullTransfer
```

DHCP — scope `client`, 10.30.0.0/24

```
Install-WindowsFeature DHCP -IncludeManagementTools
Add-DhcpServerInDC -DnsName nw-srv.paris.local -IPAddress 10.10.0.11
Set-ItemProperty -Path HKLM:\SOFTWARE\Microsoft\ServerManager\Roles\12 -Name ConfigurationState -Value 2

Add-DhcpServerv4Scope -Name "client" -StartRange 10.30.0.100 -EndRange 10.30.0.200 `
    -SubnetMask 255.255.255.0 -State Active

Add-DhcpServerv4ExclusionRange -ScopeId 10.30.0.0 -StartRange 10.30.0.100 -EndRange 10.30.0.150

# 13 days, 13 hours, 13 minutes
Set-DhcpServerv4Scope -ScopeId 10.30.0.0 -LeaseDuration (New-TimeSpan -Days 13 -Hours 13 -Minutes 13)
```

```
Set-DhcpServerv4OptionValue -ScopeId 10.30.0.0 -Router 10.30.0.1 `
-DnsServer 10.10.0.10,10.10.0.11 -DnsDomain paris.local
```

The effective usable pool is **10.30.0.151–10.30.0.200 (50 addresses)** because the exclusion sits inside the lease range.

4.3 PARIS-ROUTER — paris-router.paris.local

Four interfaces. Name them before addressing.

```
Rename-NetAdapter -Name "Ethernet0" -NewName "CORE"
Rename-NetAdapter -Name "Ethernet1" -NewName "DIST"
Rename-NetAdapter -Name "Ethernet2" -NewName "CLIENT"
Rename-NetAdapter -Name "Ethernet3" -NewName "WAN-LA"

New-NetIPAddress -InterfaceAlias "CORE" -IPAddress 10.10.0.1 -PrefixLength 24
New-NetIPAddress -InterfaceAlias "DIST" -IPAddress 10.20.0.1 -PrefixLength 24
New-NetIPAddress -InterfaceAlias "CLIENT" -IPAddress 10.30.0.1 -PrefixLength 24
New-NetIPAddress -InterfaceAlias "WAN-LA" -IPAddress 20.0.0.2 -PrefixLength 24 -DefaultGateway 20.0.0.1

Set-DnsClientServerAddress -InterfaceAlias "CORE" -ServerAddresses 10.10.0.10,10.10.0.11
Rename-Computer -NewName PARIS-ROUTER -Restart
Add-Computer -DomainName paris.local -Credential (Get-Credential PARIS\Administrator) -Restart
```

Routing and DHCP relay

```
Install-WindowsFeature RemoteAccess,Routing,RSAT-RemoteAccess-PowerShell -IncludeManagementTools
Install-RemoteAccess -VpnType RoutingOnly

# Static routes toward Lyon via the LA exchange
New-NetRoute -DestinationPrefix 10.40.0.0/24 -InterfaceAlias "WAN-LA" -NextHop 20.0.0.1
New-NetRoute -DestinationPrefix 30.0.0.0/24 -InterfaceAlias "WAN-LA" -NextHop 20.0.0.1

# DHCP relay so the CLIENT subnet reaches NW-SRV
netsh routing ip relay install
netsh routing ip relay add dhcpserver 10.10.0.11
netsh routing ip relay add interface name="CLIENT"
netsh routing ip relay set interface name="CLIENT" relaymode=enable maxhop=4 minsecs=4
```

Reverse proxy (IIS + ARR)

Uses the MSIs supplied on the Administrator Desktop.

```
Install-WindowsFeature Web-Server,Web-Mgmt-Console -IncludeManagementTools
Start-Process msixexec.exe -Wait -ArgumentList '/i "C:\Users\Administrator\Desktop\rewrite_amd64_en-US.msi" /qn'
Start-Process msixexec.exe -Wait -ArgumentList '/i "C:\Users\Administrator\Desktop\requestRouter_amd64.msi" /qn'

# Enable ARR proxy
Set-WebConfigurationProperty -PSPath 'MACHINE/WEBROOT/APPHOST' -Filter "system.webServer/proxy" -Name "enabled" -Value "True"

# Inbound rule: external.* -> WEB-SRV
Add-WebConfigurationProperty -PSPath 'MACHINE/WEBROOT/APPHOST' -Filter "system.webServer/rewrite/globalRules" -Name "." `
-Value @{name='ReverseProxyExternal';patternSyntax='Regular Expressions';stopProcessing='True'}
```

```
Set-WebConfigurationProperty -PSPath 'MACHINE/WEBROOT/APPHOST' -Filter "system.webServer/rewrite/globalRules/rule[@name='ReverseProxyExternal']/match" -Name "url" -Value "(.*)"
```

```
Set-WebConfigurationProperty -PSPath 'MACHINE/WEBROOT/APPHOST' -Filter "system.webServer/rewrite/globalRules/rule[@name='ReverseProxyExternal']/action" -Name "type" -Value "Rewrite"
```

```
Set-WebConfigurationProperty -PSPath 'MACHINE/WEBROOT/APPHOST' -Filter "system.webServer/rewrite/globalRules/rule[@name='ReverseProxyExternal']/action" -Name "url" -Value "http://10.20.0.11/{R:1}"
```

VPN (Phase 10, after LYON-ROUTER routing is live)

```
Add-VpnS2SInterface -Name "IKEv2" -Protocol IKEv2 `
-Destination 30.0.0.2 `
-AuthenticationMethod PSKOnly -SharedSecret "AllTooWell113@" `
-IPv4Subnet @"(10.40.0.0/24:100)" `
-Persistent -PassThru
```

```
Set-VpnS2SInterface -Name "IKEv2" -InitiateConfigPayload $false
```

```
Connect-VpnS2SInterface -Name "IKEv2"
```

4.4 LA-ROUTER — workgroup ISP (20.0.0.1, 30.0.0.1)

```
Rename-NetAdapter -Name "Ethernet0" -NewName "TO-PARIS"
```

```
Rename-NetAdapter -Name "Ethernet1" -NewName "TO-LYON"
```

```
New-NetIPAddress -InterfaceAlias "TO-PARIS" -IPAddress 20.0.0.1 -PrefixLength 24
```

```
New-NetIPAddress -InterfaceAlias "TO-LYON" -IPAddress 30.0.0.1 -PrefixLength 24
```

```
Set-DnsClientServerAddress -InterfaceAlias "TO-PARIS" -ServerAddresses 127.0.0.1
```



```
Rename-Computer -NewName "LA-Router" -Restart
```

```
Add-Computer -WorkgroupName "ISP" -Restart # DO NOT domain-join
```

```
Install-WindowsFeature RemoteAccess,Routing,DNS -IncludeManagementTools
```

```
Install-RemoteAccess -VpnType RoutingOnly
```



```
Add-DnsServerPrimaryZone -Name "paris.com" -ZoneFile "paris.com.dns"
```

```
Add-DnsServerResourceRecordA -ZoneName "paris.com" -Name "external" -IPv4Address 20.0.0.2
```

Verify that <http://external.paris.com> resolves and loads in Edge on LA-ROUTER. This proves the PARIS-ROUTER reverse proxy chain end to end.

4.5 FILE-SRV — file-srv.paris.local (10.20.0.10)

```
$if = (Get-NetAdapter | Where-Object Status -eq 'Up').Name
```

```
New-NetIPAddress -InterfaceAlias $if -IPAddress 10.20.0.10 -PrefixLength 24 -DefaultGateway 10.20.0.1
```

```
Set-DnsClientServerAddress -InterfaceAlias $if -ServerAddresses 10.10.0.10,10.10.0.11
```

```
Rename-Computer -NewName FILE-SRV -Restart
```

```
Add-Computer -DomainName paris.local -Credential (Get-Credential PARIS\Administrator) -Restart
```



```
Install-WindowsFeature FS-FileServer,FS-Resource-Manager,FS-DFS-Namespaces,FS-DFS-Replication, `
RSAT-DFS-Mgmt-Con,RSAT-FSRM-Mgmt -IncludeManagementTools
```

Group shares in C:

```
$root = "C:\GroupShare"
"CommonShare", "MKT", "SALES", "TECH", "HR" | ForEach-Object { New-Item -Path "$root\$_" -ItemType Directory -Force }

New-SmbShare -Name "CommonShare" -Path "$root\CommonShare" -FullAccess "Everyone"
"MKT", "SALES", "TECH", "HR" | ForEach-Object { New-SmbShare -Name $_ -Path "$root\$_" -FullAccess "Everyone" }

function Set-DeptAcl($Path,$Group){
    icacls $Path /inheritance:r | Out-Null
    icacls $Path /grant "SYSTEM:(OI)(CI)F" "BUILTIN\Administrators:(OI)(CI)F" "PARIS\Group:(OI)(CI)M" | Out-Null
}
Set-DeptAcl "$root\MKT" "MKT"
Set-DeptAcl "$root\SALES" "SALES"
Set-DeptAcl "$root\TECH" "TECH"
Set-DeptAcl "$root\HR" "HR"

# CommonShare: all users read/write EXCEPT HR -> grant each non-HR group, then explicit DENY for HR
icacls "$root\CommonShare" /inheritance:r
icacls "$root\CommonShare" /grant "SYSTEM:(OI)(CI)F" "BUILTIN\Administrators:(OI)(CI)F"
icacls "$root\CommonShare" /grant "PARIS\MKT:(OI)(CI)M" "PARIS\SALES:(OI)(CI)M" "PARIS\TECH:(OI)(CI)M"
icacls "$root\CommonShare" /deny "PARIS\HR:(OI)(CI)M"
```

Personal drives, quota and file screen

```
New-Item -Path "C:\UserShare" -ItemType Directory -Force
New-SmbShare -Name "UserShare$" -Path "C:\UserShare" -FullAccess "Everyone"
icacls "C:\UserShare" /grant "PARIS\Domain Users:(CI)(RX,AD)"

# 50 MB hard quota auto-applied to all current AND future subfolders
New-FsrmQuotaTemplate -Name "Personal-50MB" -Size 50MB -SoftLimit:$false
New-FsrmAutoQuota -Path "C:\UserShare" -Template "Personal-50MB"

# Block executables
New-FsrmFileScreen -Path "C:\UserShare" -Template "Block Executable Files" -Active
```

The home-folder attribute that maps the personal drive to τ :\ is set per user in the creation script — see section 5.3.

DFS namespace and replication

```
New-Item -Path "C:\DFSRoots\CSDrive" -ItemType Directory -Force
New-SmbShare -Name "CSDrive" -Path "C:\DFSRoots\CSDrive" -FullAccess "Everyone"
New-DfsnRoot -TargetPath "\\file-srv.paris.local\CSDrive" -Path "\\paris.local\CSDrive" -Type DomainV2

"CommonShare", "MKT", "SALES", "TECH", "HR" | ForEach-Object {
    New-DfsnFolderTarget -Path "\\paris.local\CSDrive\$_" -TargetPath "\\file-srv.paris.local\$_"
    New-DfsnFolderTarget -Path "\\paris.local\CSDrive\$_" -TargetPath "\\web-srv.paris.local\$_"
}

New-DfsReplicationGroup -GroupName "CSDrive-RG"
"CCommonShare", "MKT", "SALES", "TECH", "HR" | ForEach-Object {
    New-DfsReplicatedFolder -GroupName "CSDrive-RG" -FolderName $_
```

```

}
Add-DfsrMember -GroupName "CSDrive-RG" -ComputerName FILE-SRV,WEB-SRV
Add-DfsrConnection -GroupName "CSDrive-RG" -SourceComputerName FILE-SRV -DestinationComputerName WEB-SRV

"CommonShare","MKT","SALES","TECH","HR" | ForEach-Object {
    Set-DfsrMembership -GroupName "CSDrive-RG" -FolderName $_ -ComputerName FILE-SRV `
        -ContentPath "C:\GroupShare\$_" -PrimaryMember $true -Force
    Set-DfsrMembership -GroupName "CSDrive-RG" -FolderName $_ -ComputerName WEB-SRV `
        -ContentPath "C:\GroupShare\$_" -Force
}

```

RemoteApp (notepad.exe)

Server Manager GUI: Manage > Add Roles and Features > **Remote Desktop Services installation** > Quick Start > Session-based desktop deployment > select FILE-SRV.

```

New-RDSessionDeployment -ConnectionBroker "file-srv.paris.local" `
    -WebAccessServer "file-srv.paris.local" -SessionHost "file-srv.paris.local"

New-RDSessionCollection -CollectionName "AppCollection" `
    -SessionHost "file-srv.paris.local" -ConnectionBroker "file-srv.paris.local"

New-RDRemoteApp -Alias "notepad" -DisplayName "Notepad" `
    -FilePath "C:\Windows\System32\notepad.exe" `
    -ShowInWebAccess 1 -CollectionName "AppCollection" -ConnectionBroker "file-srv.paris.local"

```

A self-signed certificate is explicitly acceptable here — HTTPS compliance is not required for RemoteApp.

4.6 WEB-SRV — web-srv.paris.local (10.20.0.11)

```

$if = (Get-NetAdapter | Where-Object Status -eq 'Up').Name
New-NetIPAddress -InterfaceAlias $if -IPAddress 10.20.0.11 -PrefixLength 24 -DefaultGateway 10.20.0.1
Set-DnsClientServerAddress -InterfaceAlias $if -ServerAddresses 10.10.0.10,10.10.0.11
Rename-Computer -NewName WEB-SRV -Restart
Add-Computer -DomainName paris.local -Credential (Get-Credential PARIS\Administrator) -Restart

```

PKI first

```

Install-WindowsFeature ADCS-Cert-Authority,ADCS-Web-Enrollment,RSAT-ADCS,RSAT-ADCS-Mgmt -IncludeManagementTools

Install-AdcsCertificationAuthority -CAType EnterpriseRootCA `
    -CACommonName "paris-WEB-SRV-CA" `
    -KeyLength 4096 -HashAlgorithmName SHA256 `
    -ValidityPeriod Years -ValidityPeriodUnits 10 -Force

Install-AdcsWebEnrollment -Force

```

An **Enterprise** Root CA auto-publishes its root certificate to the domain-wide Trusted Root and NTAAuth stores. This is what satisfies “distribute them domain wide” and “no certificate errors in the Edge web browser”.

Issue the web certificates

```
Get-Certificate -Template "WebServer" -SubjectName "CN=www.paris.local" `
-DnsName "www.paris.local" -CertStoreLocation Cert:\LocalMachine\My
Get-Certificate -Template "WebServer" -SubjectName "CN=help.paris.local" `
-DnsName "help.paris.local" -CertStoreLocation Cert:\LocalMachine\My
```

The WebServer template must first be published and its enrolment permissions granted — see section 5.7.

IIS and virtual hosts

```
Install-WindowsFeature Web-Server,Web-Mgmt-Console,Web-Http-Redirect -IncludeManagementTools
Start-Process msixexec.exe -Wait -ArgumentList '/i "C:\Users\Administrator\Desktop\rewrite_amd64_en-US.msi" /qn'

$tpl = "C:\Users\Administrator\Desktop\HTML templates"
New-Item C:\inetpub\www,C:\inetpub\help,C:\inetpub\external -ItemType Directory -Force
Copy-Item "$tpl\internal.html" C:\inetpub\www\index.html
Copy-Item "$tpl\help.html" C:\inetpub\help\index.html
Copy-Item "$tpl\external.html" C:\inetpub\external\index.html

Remove-Website -Name "Default Web Site"
New-Website -Name "www" -HostHeader "www.paris.local" -Port 80 -PhysicalPath C:\inetpub\www
New-Website -Name "help" -HostHeader "help.paris.local" -Port 80 -PhysicalPath C:\inetpub\help
New-Website -Name "external" -HostHeader "external.paris.local" -Port 80 -PhysicalPath C:\inetpub\external

$wwwCert = (Get-ChildItem Cert:\LocalMachine\My | Where-Object Subject -like "*www.paris.local*")[0]
$helpCert = (Get-ChildItem Cert:\LocalMachine\My | Where-Object Subject -like "*help.paris.local*")[0]
New-WebBinding -Name "www" -Protocol https -Port 443 -HostHeader "www.paris.local" -SslFlags 1
New-WebBinding -Name "help" -Protocol https -Port 443 -HostHeader "help.paris.local" -SslFlags 1
New-Item -Path "IIS:\SslBindings\!443!www.paris.local" -Value $wwwCert -SSLFlags 1
New-Item -Path "IIS:\SslBindings\!443!help.paris.local" -Value $helpCert -SSLFlags 1
```

HTTP to HTTPS redirect for www and help only

```
foreach ($site in "www","help") {
    Add-WebConfigurationProperty -PSPath "IIS:\Sites\$site" -Filter "system.webServer/rewrite/rules" -Name "." `
    -Value @{name='HTTPtoHTTPS';stopProcessing='True'}
    Set-WebConfigurationProperty -PSPath "IIS:\Sites\$site" -Filter "system.webServer/rewrite/rules/rule[@name='HTTPtoHTTPS']/match" -Name "url" -Value "(.*)"
    Add-WebConfigurationProperty -PSPath "IIS:\Sites\$site" -Filter "system.webServer/rewrite/rules/rule[@name='HTTPtoHTTPS']/conditions" -Name "." `
    -Value @{input='{HTTPS}';pattern='^OFF$'}
    Set-WebConfigurationProperty -PSPath "IIS:\Sites\$site" -Filter "system.webServer/rewrite/rules/rule[@name='HTTPtoHTTPS']/action" -Name "type" -Value
    "Redirect"
    Set-WebConfigurationProperty -PSPath "IIS:\Sites\$site" -Filter "system.webServer/rewrite/rules/rule[@name='HTTPtoHTTPS']/action" -Name "url" -Value
    "https://{HTTP_HOST}/{R:1}"
    Set-WebConfigurationProperty -PSPath "IIS:\Sites\$site" -Filter "system.webServer/rewrite/rules/rule[@name='HTTPtoHTTPS']/action" -Name "redirectType" -Value
    "Permanent"
}
```

DFS-R partner shares

```
Install-WindowsFeature FS-FileServer,FS-DFS-Replication -IncludeManagementTools
"CommonShare","MKT","SALES","TECH","HR" | ForEach-Object {
    New-Item "C:\GroupShare\$_" -ItemType Directory -Force
}
```

```
New-SmbShare -Name $_ -Path "C:\GroupShare\$_" -FullAccess "Everyone" -ErrorAction SilentlyContinue
}
```

4.7 LYON-ROUTER — lyon-router.lyon.paris.local

```
Rename-NetAdapter -Name "Ethernet0" -NewName "LAN-LYON"
Rename-NetAdapter -Name "Ethernet1" -NewName "WAN-LA"
New-NetIPAddress -InterfaceAlias "LAN-LYON" -IPAddress 10.40.0.1 -PrefixLength 24
New-NetIPAddress -InterfaceAlias "WAN-LA" -IPAddress 30.0.0.2 -PrefixLength 24 -DefaultGateway 30.0.0.1
Set-DnsClientServerAddress -InterfaceAlias "LAN-LYON" -ServerAddresses 10.40.0.10
Rename-Computer -NewName LYON-ROUTER -Restart

Install-WindowsFeature RemoteAccess, Routing, RSAT-RemoteAccess-PowerShell -IncludeManagementTools
Install-RemoteAccess -VpnType VpnS2S

New-NetRoute -DestinationPrefix 10.10.0.0/24 -InterfaceAlias "WAN-LA" -NextHop 30.0.0.1
New-NetRoute -DestinationPrefix 10.20.0.0/24 -InterfaceAlias "WAN-LA" -NextHop 30.0.0.1
New-NetRoute -DestinationPrefix 10.30.0.0/24 -InterfaceAlias "WAN-LA" -NextHop 30.0.0.1

Add-VpnS2SInterface -Name "IKEv2" -Protocol IKEv2 -Destination 20.0.0.2 `
  -AuthenticationMethod PSKOnly -SharedSecret "AllTooWell13@" `
  -IPv4Subnet @("10.10.0.0/24:100", "10.20.0.0/24:100", "10.30.0.0/24:100") -Persistent
Connect-VpnS2SInterface -Name "IKEv2"
```

After DC2 exists

```
Add-Computer -DomainName lyon.paris.local -Credential (Get-Credential LYON\Administrator) -Restart

Install-WindowsFeature DHCP -IncludeManagementTools
Add-DhcpServerInDC -DnsName lyon-router.lyon.paris.local -IPAddress 10.40.0.1

Add-DhcpServerv4Scope -Name "client" -StartRange 10.40.0.100 -EndRange 10.40.0.200 -SubnetMask 255.255.255.0 -State Active
Add-DhcpServerv4ExclusionRange -ScopeId 10.40.0.0 -StartRange 10.40.0.100 -EndRange 10.40.0.150
Set-DhcpServerv4Scope -ScopeId 10.40.0.0 -LeaseDuration (New-TimeSpan -Days 13 -Hours 13 -Minutes 13)
Set-DhcpServerv4OptionValue -ScopeId 10.40.0.0 -Router 10.40.0.1 -DnsServer 10.40.0.10 -DnsDomain lyon.paris.local
```

4.8 DC2 — dc2.lyon.paris.local (10.40.0.10)

```
$if = (Get-NetAdapter | Where-Object Status -eq 'Up').Name
New-NetIPAddress -InterfaceAlias $if -IPAddress 10.40.0.10 -PrefixLength 24 -DefaultGateway 10.40.0.1
Set-DnsClientServerAddress -InterfaceAlias $if -ServerAddresses 10.10.0.10 # parent DNS during promotion
Rename-Computer -NewName DC2 -Restart

Install-WindowsFeature AD-Domain-Services, DNS -IncludeManagementTools
Import-Module ADDSDeployment
Install-ADDSDomain `
  -NewDomainName "lyon" `
  -ParentDomainName "paris.local" `
  -NewDomainNetbiosName "LYON" `
  -DomainType ChildDomain `
  -InstallDns:$true -CreateDnsDelegation:$true `
  -Credential (Get-Credential PARIS\Administrator) `
```

```
-SafeModeAdministratorPassword (ConvertTo-SecureString "Skill39@Lyon" -AsPlainText -Force) `
-DatabasePath "C:\Windows\NTDS" -LogPath "C:\Windows\NTDS" -SysvolPath "C:\Windows\SYSVOL" -Force
```

After reboot, repoint DNS to itself with the parent as secondary:

```
Set-DnsClientServerAddress -InterfaceAlias $if -ServerAddresses 10.40.0.10,10.10.0.10
Add-DnsServerConditionalForwarderZone -Name "paris.local" -MasterServers 10.10.0.10 -ReplicationScope Domain
```

4.9 Clients

```
# WIN-CLIENT1
Rename-Computer -NewName WIN-CLIENT1 -Restart
Add-Computer -DomainName paris.local -Credential (Get-Credential PARIS\Administrator) -Restart

# WIN-CLIENT2
Rename-Computer -NewName WIN-CLIENT2 -Restart
Add-Computer -DomainName lyon.paris.local -Credential (Get-Credential LYON\Administrator) -Restart
```

Both clients must obtain addresses via DHCP. Do not set static IPs.

5. Configuration Details

5.1 AD DS Structure — paris.local

OU (path)	Global security group	Members
OU=MKT,DC=paris,DC=local	MKT	mkt1–mkt20
OU=SALES,DC=paris,DC=local	SALES	sales1–sales20
OU=TECH,DC=paris,DC=local	TECH	tech1–tech20
OU=HR,DC=paris,DC=local	HR	hr1–hr20

lyon.paris.local: OU REMOTE, group REMOTE, users remote1–remote20.

```
"MKT","SALES","TECH","HR" | ForEach-Object {
    New-ADOrganizationalUnit -Name $_ -Path "DC=paris,DC=local" -ProtectedFromAccidentalDeletion $false
    New-ADGroup -Name $_ -GroupScope Global -GroupCategory Security -Path "OU=$_,DC=paris,DC=local"
}
```

5.2 Passwords

Scope	Password
-------	----------

Scope	Password
Server / client default	Skill139@Lyon
mkt / sales / tech / remote users	Skill139@Lyon
hr users	Skill139@LyonSkill139@Lyon — 22 characters, satisfies the 16-character FGPP

5.3 C:_user.ps1

```
<# C:\create_user.ps1 - Usage: C:\create_user.ps1 -count 5 #>
[CmdletBinding()]
param(
    [Parameter(Mandatory=$true)][int]$count
)

Import-Module ActiveDirectory

$departments = @(
    @{ Prefix='mkt'; OU='MKT'; Group='MKT'; Password='Skill139@Lyon' }
    @{ Prefix='sales'; OU='SALES'; Group='SALES'; Password='Skill139@Lyon' }
    @{ Prefix='tech'; OU='TECH'; Group='TECH'; Password='Skill139@Lyon' }
    @{ Prefix='hr'; OU='HR'; Group='HR'; Password='Skill139@LyonSkill139@Lyon' }
)

foreach ($d in $departments) {
    $ouPath = "OU= $($d.OU),DC=paris,DC=local"
    $secPwd = ConvertTo-SecureString $d.Password -AsPlainText -Force

    for ($i = 1; $i -le $count; $i++) {
        $sam = "$($d.Prefix)$i"

        if (Get-ADUser -Filter "SamAccountName -eq '$sam'" -ErrorAction SilentlyContinue) {
            Write-Host "SKIP : $sam already exists"
            continue
        }

        New-ADUser -Name $sam -SamAccountName $sam `
            -UserPrincipalName "$sam@paris.local" `
            -Path $ouPath `
            -AccountPassword $secPwd `
            -Enabled $true `
            -ChangePasswordAtLogon $false `
            -PasswordNeverExpires $false `
            -HomeDrive "T:" `
            -HomeDirectory "\\file-srv.paris.local\UserShare$\$sam"

        Add-ADGroupMember -Identity $d.Group -Members $sam
        Write-Host "CREATE: $sam -> $($d.OU)"
    }
}
```

-HomeDrive T: combined with -HomeDirectory auto-maps the personal drive to T:\ and creates the folder with the correct owner ACL. Run the full set with C:\create_user.ps1 -count 20.

5.4 Fine-Grained Password Policy (HR)

```
New-ADFineGrainedPasswordPolicy -Name "HR-PSO" `
  -Precedence 10 `
  -MinPasswordLength 16 `
  -ComplexityEnabled $true `
  -PasswordHistoryCount 24 `
  -MinPasswordAge "1.00:00:00" -MaxPasswordAge "42.00:00:00" `
  -LockoutThreshold 5 -LockoutDuration "0.00:30:00" -LockoutObservationWindow "0.00:30:00"

Add-ADFineGrainedPasswordPolicySubject -Identity "HR-PSO" -Subjects "HR"
```

5.5 Group Policy Objects

GPO	Link target	Setting	Implementation
Desktop	DC=paris,DC=local	Logon banner text and caption	Computer Config > Policies > Security Options > Interactive logon message
Desktop	"	Env var TheErasTour = 2024	Computer Config > Preferences > Environment > System variable
Desktop	"	Disable local Administrator	Security Options > Accounts: Administrator account status = Disabled
Desktop	"	No LM hash stored	HKLM\SYSTEM\CurrentControlSet\Control\Lsa\NoLMHash = 1
Desktop	"	Updates Friday 13:00	Windows Update > Configure Automatic Updates = 4, day 6, time 13
TECH	OU=TECH	powershell.exe at logon	User Config > Preferences > Registry > HKCU\...\Run\TECHShell
NOTECH	OU=MKT, OU=SALES, OU=HR (three links)	Disable registry editing, cmd, Run, PowerShell; File History off	User and Computer Config — see script

```
Import-Module GroupPolicy
$banner = "Welcome to Lyon! Only authorised personnel allowed to access. Should you try to break in, I knew you were trouble"

# ---- Desktop GPO ----
New-GPO -Name "Desktop" | New-GPLink -Target "DC=paris,DC=local" -LinkEnabled Yes

Set-GPRegistryValue -Name "Desktop" -Key "HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\System" `
  -ValueName "legalnoticecaption" -Type String -Value "ErasTour24"
Set-GPRegistryValue -Name "Desktop" -Key "HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\System" `
```

```

-ValueName "legalnoticetext" -Type String -Value $banner

Set-GPRegistryValue -Name "Desktop" -Key "HKLM\SYSTEM\CurrentControlSet\Control\Lsa" `
-ValueName "NoLMHash" -Type DWord -Value 1

Set-GPRegistryValue -Name "Desktop" -Key "HKLM\Software\Policies\Microsoft\Windows\WindowsUpdate\AU" `
-ValueName "NoAutoUpdate" -Type DWord -Value 0
Set-GPRegistryValue -Name "Desktop" -Key "HKLM\Software\Policies\Microsoft\Windows\WindowsUpdate\AU" `
-ValueName "AUOptions" -Type DWord -Value 4
Set-GPRegistryValue -Name "Desktop" -Key "HKLM\Software\Policies\Microsoft\Windows\WindowsUpdate\AU" `
-ValueName "ScheduledInstallDay" -Type DWord -Value 6 # 6 = Friday
Set-GPRegistryValue -Name "Desktop" -Key "HKLM\Software\Policies\Microsoft\Windows\WindowsUpdate\AU" `
-ValueName "ScheduledInstallTime" -Type DWord -Value 13 # 13:00

# ---- TECH GPO ----
New-GPO -Name "TECH" | New-GPLink -Target "OU=TECH,DC=paris,DC=local" -LinkEnabled Yes
Set-GPRegistryValue -Name "TECH" -Key "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" `
-ValueName "TECHShell" -Type String -Value "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"

# ---- NOTECH GPO ----
New-GPO -Name "NOTECH"
"OU=MKT,DC=paris,DC=local", "OU=SALES,DC=paris,DC=local", "OU=HR,DC=paris,DC=local" | ForEach-Object {
    New-GPLink -Name "NOTECH" -Target $_ -LinkEnabled Yes
}
Set-GPRegistryValue -Name "NOTECH" -Key "HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\System" `
-ValueName "DisableRegistryTools" -Type DWord -Value 2
Set-GPRegistryValue -Name "NOTECH" -Key "HKCU\Software\Policies\Microsoft\Windows\System" `
-ValueName "DisableCMD" -Type DWord -Value 2
Set-GPRegistryValue -Name "NOTECH" -Key "HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer" `
-ValueName "NoRun" -Type DWord -Value 1
Set-GPRegistryValue -Name "NOTECH" -Key "HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer" `
-ValueName "DisallowRun" -Type DWord -Value 1
Set-GPRegistryValue -Name "NOTECH" -Key "HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\DisallowRun" `
-ValueName "1" -Type String -Value "powershell.exe"
Set-GPRegistryValue -Name "NOTECH" -Key "HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\DisallowRun" `
-ValueName "2" -Type String -Value "powershell_ise.exe"
Set-GPRegistryValue -Name "NOTECH" -Key "HKLM\Software\Policies\Microsoft\Windows\FileHistory" `
-ValueName "Disabled" -Type DWord -Value 1

```

Disabling the local Administrator account cannot be set with Set-GPRegistryValue. In GPMC, edit **Desktop** > Computer Configuration > Policies > Windows Settings > Security Settings > Local Policies > Security Options > **Accounts: Administrator account status** > Define > **Disabled**.

Drive mappings (User Configuration > Preferences > Drive Maps, item-level targeted)

Drive	Target	Targeted at
Z:	\\paris.local\CSDrive\CommonShare	Authenticated Users (excluding HR)
X:	\\paris.local\CSDrive\MKT	PARIS\MKT
X:	\\paris.local\CSDrive\SALES	PARIS\SALES
X:	\\paris.local\CSDrive\TECH	PARIS\TECH

Drive	Target	Targeted at
X:	\\paris.local\CSDrive\HR	PARIS\HR
T:	Home directory (set on user object)	All users

5.6 DNS Zones Summary

Server	Zone	Type	Notes
DC1	paris.local	AD-integrated primary	www, help, external, app + AD SRV records
DC1	0.10.10, 0.20.10, 0.30.10 .in-addr.arpa	Primary reverse	Transfer to 10.10.0.11
NW-SRV	all of the above	Secondary	Master 10.10.0.10
DC2	lyon.paris.local	AD-integrated primary	Delegation created in parent
LA-ROUTER	paris.com	Standalone primary	external to 20.0.0.2
Forwarders	[TO BE CONFIRMED]	—	No upstream resolver specified

5.7 Certificate Templates (WEB-SRV CA)

```
# Publish the WebServer template for issuance
certutil -SetCAtemplates +WebServer
```

GUI: certsrv.msc > Certificate Templates > New > Certificate Template to Issue > **Web Server**. Then certtmpl.msc > Web Server > Properties > Security > add **Domain Computers** > grant **Enroll**.

6. Validation & Testing

Phase	Command	Expected healthy output
1 — LA-ROUTER	Get-NetIPAddress -AddressFamily IPv4 \\\ ft IPAddress,InterfaceAlias; Resolve-DnsName external.paris.com -Server 127.0.0.1	20.0.0.1 and 30.0.0.1 present; record returns 20.0.0.2
2 — DC1 forest	dcldiag /v /c /e; Get-ADForest \\\ fl Name,ForestMode; Resolve-DnsName _ldap._tcp.dc._msdcs.paris.local - Type SRV	All tests passed; forest paris.local; SRV resolves to dc1.paris.local
3 — Routing	Get-NetRoute -DestinationPrefix 10.40.0.0/24; Test- NetConnection 20.0.0.1 -TraceRoute	Route present; ping succeeds

Phase	Command	Expected healthy output
4 — Reverse zone	Resolve-DnsName 10.20.0.11 -Type PTR	Returns web-srv.paris.local
5 — Secondary DNS	Get-DnsServerZone -ComputerName NW-SRV; Resolve-DnsName www.paris.local -Server 10.10.0.11	Zones show type Secondary, not Shutdown; resolves 10.20.0.11
5 — DHCP	Get-DhcpServerInDC; Get-DhcpServerv4Scope; Get-DhcpServerv4ExclusionRange -ScopeId 10.30.0.0; Get-DhcpServerv4Lease -ScopeId 10.30.0.0	NW-SRV authorised; scope client Active, lease 13.13:13:00; exclusion 100–150; leases issued in 151–200
6a — Shares	Get-SmbShare; Get-FsrmAutoQuota -Path C:\UserShare; Get-FsrmFileScreen -Path C:\UserShare	Five group shares plus UserShare\$; 50 MB hard quota; executable screen Active
6a — ACL	icacls C:\GroupShare\CommonShare	PARIS\HR:(OI)(CI)(DENY)(M) present
6b — PKI	certutil -ping; certutil -verifystore -enterprise Root; Get-CACrldDistributionPoint	CA responds; root cert in enterprise store; CDP reachable
6b — IIS	Invoke-WebRequest http://www.paris.local -MaximumRedirection 0; Invoke-WebRequest https://www.paris.local	HTTP returns 301 to HTTPS; HTTPS returns 200 with no certificate warning
7 — DFS	Get-DfsnRoot; Get-DfsnFolderTarget -Path \\paris.local\CSDrive\MKT; dfsrdiag ReplicationState /member:FILE-SRV; Get-DfsrBacklog -SourceComputerName FILE-SRV -DestinationComputerName WEB-SRV -GroupName CSDrive-RG -FolderName MKT	Root Online; two targets; state Normal; backlog 0
8 — Users / policy	Get-ADUser -Filter * -SearchBase "OU=MKT,DC=paris,DC=local" \ Measure-Object; Get-ADFineGrainedPasswordPolicy HR-PSO; Get-ADUserResultantPasswordPolicy hr1	20 users; MinPasswordLength 16; hr1 resolves to HR-PSO
10 — VPN	Get-VpnS2SInterface -Name IKEv2 \ f1 ConnectionState,Persistent; Get-NetIPsecMainModeSA	Connected, Persistent True; SA present
11 — DC2	Get-ADDomain lyon.paris.local; repadmin /replsummary; repadmin /showrepl DC2; nltest /dsgetdc:lyon.paris.local	Domain exists; zero fails in replsummary; all contexts Successful
12 — Lyon DHCP	Get-DhcpServerv4Scope -ComputerName lyon-router	Scope client Active, lease 13.13:13:00
13 — RemoteApp	Get-RDRemoteApp -CollectionName AppCollection; from WIN-CLIENT2 open https://file-srv.paris.local/RDWeb	Notepad listed; launches from both client networks
14 — Clients	gpresult /r /scope:computer; gpresult /r /scope:user; Get-PSDrive -PSProvider FileSystem	Desktop in Applied GPOs; TECH or NOTECH per user OU; T, Z, and X: mapped

Phase	Command	Expected healthy output
14 — Banner	Log off and log on WIN-CLIENT1	Banner text appears before the credential prompt
14 — NOTECH	As sales1, press Win+R; run regedit	Run dialog blocked; registry editing blocked
15 — Cross-site	From WIN-CLIENT2: Test-NetConnection file-srv.paris.local -Port 445; Test-NetConnection 10.20.0.11 -Port 443	TcpTestSucceeded : True on both
16 — Ansible	cd /opt/ansible && ansible-playbook manage-shares.yaml; git log --oneline -1	Playbook completes with no credential prompt; last commit reads final commit

7. Gaps, Conflicts & Risks

7.1 Conflicts Between the Two Attachments

#	Conflict	Detail	Resolution
C1	Entirely different environments	2026 topology is Shanghai/Beijing, Server 2025, DMZ/INTERNAL/CLIENT zones; 2024 instructions are Paris/Lyon/LA, Server 2022, Core/Distribution/Client layers	Build against the 2024 values. Treat the 2026 file as a future-state reference until its redactions are released
C2	OS version mismatch	2026 specifies Windows Server 2025; 2024 specifies Windows Server 2022 21H2	Confirm media. If 2025 is mandated, all cmdlets above remain valid; RDS deployment syntax and AD CS defaults shift slightly
C3	DMZ tier absent from the 2024 build	2026 shows a DMZ zone with two core servers per site; 2024 places WEB-SRV in the internal Distribution layer with the reverse proxy on PARIS-ROUTER	Do not invent a DMZ. Flag that the 2026 design will require a separate firewalled segment and re-homing of WEB-SRV
C4	Core versus GUI on NW-SRV	Body text specifies Server Core, yet the configuration described is GUI-oriented	Build Core; administer remotely with RSAT from DC1

7.2 Gaps Inside the 2024 Instructions

#	Gap	Impact	Fix
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#	Gap	Impact	Fix
G1	help and external DNS records have no IP address in the Task 1 table	Both virtual hosts unreachable internally	Use 10.20.0.11 (WEB-SRV). Marked [TO BE CONFIRMED]
G2	10.30.01 typo for PARIS-ROUTER	Invalid IPv4 address	Read as 10.30.0.1, consistent with the DHCP default gateway
G3	DHCP server is off-subnet — NW-SRV on 10.10.0.0/24 serves 10.30.0.0/24	Clients receive no address; DHCP broadcasts do not route	Mandatory: DHCP Relay Agent on the PARIS-ROUTER CLIENT interface. Not mentioned anywhere in the document
G4	Exclusion nested inside the lease range (100–150 excluded from 100–200)	Only 50 usable addresses	Confirm intent; likely deliberate
G5	No subnet masks, VLAN IDs, VM specifications, hypervisor, storage or forest functional level	Cannot size or build without assumptions	All marked [TO BE CONFIRMED]
G6	No NTP or authoritative time source defined	Kerberos failures across the VPN	Configure the PDC Emulator (DC1) against an external source using w32tm /config
G7	No DNS forwarder or internet egress defined	External name resolution fails	[TO BE CONFIRMED] — LA-ROUTER is the only plausible upstream
G8	RD Licensing server not specified	RemoteApp stops after the 120-day grace period	Deploy the RD Licensing role with CALs, or accept the grace period
G9	Credential-less Ansible execution requires Kerberos/WinRM configuration that is not described	Playbook prompts for credentials	Configure ansible_winrm_transport: kerberos, /etc/krb5.conf for PARIS.LOCAL and LYON.PARIS.LOCAL, and kinit a service account
G10	Banner text reads “Welcome to Lyon!” but the GPO is scoped to paris.local	Cosmetic and logical mismatch	Implement verbatim — it is a literal requirement

7.3 Design Risks

Risk	Severity	Description	Recommendation
Single DC per domain	High	DC1 is the only DC for paris.local; DC2 the only one for lyon.paris.local	Add a second DC per domain; at minimum take regular System State backups and verify the DSRM

Risk	Severity	Description	Recommendation
			password
PARIS-ROUTER is a quadruple single point of failure	High	Gateway for three subnets, plus VPN endpoint, reverse proxy and DHCP relay, on one VM	Split the reverse proxy onto a dedicated host; add a standby router
No backup, antivirus or monitoring anywhere	High	Neither document specifies any	Add a backup target, endpoint protection and monitoring before production
Enterprise Root CA online, single-tier, on a web server	Medium	Root CA co-located with the internet-facing content origin	Adopt a two-tier PKI: offline standalone root plus online issuing CA. Never co-locate with IIS
Everyone full control on SMB shares	Medium	Required by the Ansible task; relies solely on NTFS for protection	Acceptable only because NTFS ACLs are restrictive. Tighten share ACLs wherever automation does not demand it
Disabling the local Administrator account domain-wide	Medium	The Desktop GPO applies at the domain root	Create and test a break-glass domain admin account before linking the GPO; exclude DCs if recovery access is a concern
Pre-shared key VPN authentication	Medium	AllTooWell113@ is a static, documented PSK	Migrate to certificate-based IKEv2 using the WEB-SRV CA
Weak, shared, reused password	Medium	Skill139@Lyon covers every server, client and 60 of 80 users	Unique credentials per tier; LAPS for local accounts
All inter-site AD traffic depends on one tunnel	Medium	If IKEv2 drops, DC2 loses replication and Lyon clients lose RemoteApp	Persistent connection is already specified; add a second tunnel or backup path and monitor the interface
DFS-R with only two members and one connection	Low	Single one-way connection FILE-SRV to WEB-SRV	Make the connection bidirectional; monitor backlog
No AD Sites and Services topology	Low	Lyon clients may authenticate against DC1 over the VPN	Define Paris and Lyon sites with subnets and a site link

8. Pre-Implementation Checklist

8.1 Hardware / Platform

- Hypervisor host(s) sized for **10 VMs** — eight Windows Server, two Windows 11, one Debian. [TO BE CONFIRMED: CPU/RAM/disk per VM not specified]
- Virtual switches or port groups for **six** Layer 2 segments: 10.10.0.0/24, 10.20.0.0/24, 10.30.0.0/24, 10.40.0.0/24, 20.0.0.0/24, 30.0.0.0/24
- PARIS-ROUTER provisioned with **four vNICs**; LA-ROUTER and LYON-ROUTER with **two vNICs** each
- Datastore capacity for DFS replicated content plus c:\UserShare (80 users × 50 MB ≈ 4 GB minimum) [TO BE CONFIRMED]

8.2 Media

- Windows Server 2022 21H2 ISO (Desktop Experience and Core install options)
- Windows 11 23H2 ISO
- Debian ISO with VS Code, Ansible and Zeal offline docs pre-staged
- requestRouter_amd64.msi staged on the WEB-SRV and PARIS-ROUTER Administrator Desktop
- rewrite_amd64_en-US.msi staged on the WEB-SRV and PARIS-ROUTER Administrator Desktop
- HTML templates folder (internal.html, help.html, external.html) staged on the WEB-SRV Administrator Desktop

8.3 Licensing

- Windows Server 2022 Standard keys ×8, or Datacenter if VM density justifies it [TO BE CONFIRMED]
- Windows 11 licences ×2
- Windows Server CALs for 80 user accounts
- RDS CALs for RemoteApp beyond the 120-day grace period [TO BE CONFIRMED]

8.4 Credentials

- Local Administrator Skill139@Lyon on all servers and clients
- DSRM password for DC1 and DC2: Skill139@Lyon
- Enterprise Admin PARIS\Administrator — required for DC2 child-domain promotion and AD CS installation
- VPN credential set: user Administrator, domain PARIS, password Skill139@Lyon, PSK AllTooWell113@
- HR user password string Skill139@LyonSkill139@Lyon
- Break-glass domain admin account created and tested **before** the Desktop GPO is linked
- Ansible service account and Kerberos keytab for credential-less WinRM [TO BE CONFIRMED]

8.5 Design Decisions to Confirm Before Day One

- IP address for the DNS records help and external
- Subnet masks and any VLAN tagging
- Forest and domain functional level
- AD Sites and site-link schedule
- Upstream DNS forwarder and internet egress path

- NTP source for the PDC Emulator
- Time zone
- Whether the 2026 Server 2025 / Shanghai–Beijing / DMZ design supersedes this build

8.6 Change Approvals

- Approval to create forest `paris.local` and child domain `lyon.paris.local`
- Approval for the IKEv2 pre-shared-key tunnel across the LA exchange
- Approval for the domain-wide GPO disabling local Administrator accounts
- Approval for `Everyone - Full Control` share permissions (NTFS-enforced)
- Maintenance window covering Phases 1–16, with rollback snapshots taken at Phases 2, 6, 10 and 11